Unknown-Aware LLM-generated Texts Detection

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Abstract

As Large Language Models continue to evolve and new models with diverse writing styles are released, detecting machine-generated text is becoming increasingly difficult. Existing approaches often frame detection as binary classification between human-written and machine-generated text, which can fail in open-set settings where text may come from unseen LLM generators. We build upon an OOD-based detection framework (Zeng et al., 2025) and propose HOUND (Human or Unknown Detection), an unknown-aware LLM-generated text detection framework using a cascaded pipeline built on StyleDistance embeddings (Patel et al., 2025). HOUND first separates human-written from LLM-generated text, then applies an energy-based unknown gate to reject unseen generators, and finally attributes accepted in-distribution samples to one of seven known LLM generators. On RAID with held-out generators, our best abstracts-only cascade achieves 72% final accuracy, 97.03% human recall, 61.79% unknown detection rate, and 93.60% known-LLM attribution accuracy after the gate. A domain-balanced version is more realistic but harder, achieving 62% final accuracy, 96.77% human recall, 45.87% unknown detection rate, and 91.96% known-LLM attribution accuracy after the gate. These results show that energy-based gating is useful for unknown-aware LLM attribution, while unknown-generator rejection remains challenging under domain shift.

1 Introduction

Large language models like ChatGPT, Gemini, DeepSeek, and Claude have become increasingly popular in recent years. As a result, LLM-generated text has become common across many domains. Without reliable detection frameworks, this can lead to misuse such as impersonation, academic dishonesty, and scaled phishing operations (Marchal et al., 2024). A further challenge is that online text is often not fully human-written or machine-written, but a mixture of human-edited and machine-generated text. Since new models are released frequently, detection systems must also generalize to unseen or novel generators.

Existing detection models are often zero-shot, such as DetectLLM (Su et al., 2023), or training-based, such as GPTZero (Adam et al., 2026). Most training-based methods frame detection as a binary decision between human-written and machine-written text. However, *Human Texts Are Outliers: Detecting LLM-generated Texts via Out-of-distribution Detection* (Zeng et al., 2025) argues that binary classification is inadequate because human-written text varies widely across styles and domains, while machine-generated text is often more statistically compact. HTAO therefore reframes detection as an out-of-distribution problem, treating machine-written text as in-distribution and human-written text as out-of-distribution.

We argue that LLM-generated text detection should also be treated as an *unknown-aware* problem: a classifier needs to recognize not only human text, but also text from LLMs it has never seen. We introduce **HOUND** (Human or Unknown Detection), a cascaded framework that classifies input text as human-written, from a known LLM, or from an unknown LLM generator. HOUND uses StyleDistance embeddings (Patel et al., 2025), a human-vs-LLM classifier, an energy-based unknown gate, and a known-LLM attribution classifier. We chose a cascade rather than a single three-way classifier because human detection and known-LLM attribution were already strong signals, while unknown rejection required a separate thresholded decision.

We evaluate HOUND on the RAID benchmark (Dugan et al., 2024), which contains human and multi-generator LLM text across multiple domains and adversarial perturbations. On a domain-balanced RAID setting with held-out gen-

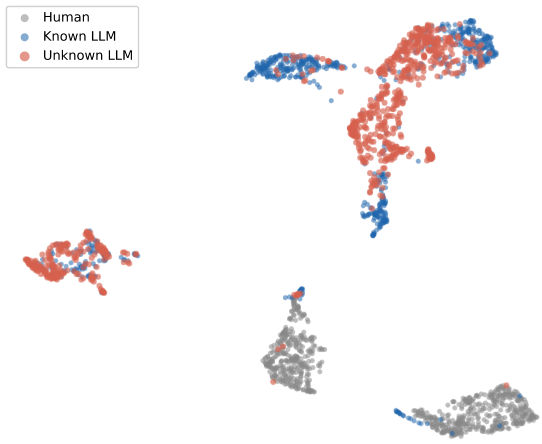

Figure 1: UMAP projection of StyleDistance embeddings on RAID (800 samples per class). Human text (gray) separates clearly from LLM text, supporting the strong human-vs-LLM performance of our cascade. Known LLMs (blue) and unknown LLMs (red), however, substantially overlap in style space, motivating the energy-based gate for unknown-aware detection.

erators, HOUND achieves 96.8% recall on human text and 92% known-LLM attribution accuracy on samples that pass the gate, while detecting 45.9% of unknown-generator text at 71% precision. These results show that cascaded unknown-aware detection is promising, but unknown-generator rejection remains the main challenge.

Our contributions are as follows:

- We introduce HOUND, a cascaded unknown-aware framework for LLM-generated text detection that jointly performs human-vs-LLM separation, unknown-generator rejection, and known-LLM family attribution.
- We build HOUND using StyleDistance embeddings and compare several unknown-rejection signals, including centroid distance, DeepSVDD-style distance, energy scoring, and confidence-based scores.
- We evaluate HOUND on RAID with held-out generators and identify the unknown-precision/known-recall tradeoff as the main open problem for unknown-aware text detection.

2 Related Works

2.1 LLM-Generated Text Detection

Prior work on LLM-generated text detection includes both zero-shot methods and training-based detectors. Zero-shot approaches such as DetectGPT and DetectLLM use properties of language model likelihoods or perturbations to identify generated text (Su et al., 2023). Training-based approaches learn a supervised boundary between human and machine-written text. These methods can perform well when training and testing distributions match, but they are less reliable when new generators, new domains, or adversarially edited text appear at test time.

2.2 Human Texts Are Outliers

HTAO reframes LLM-generated text detection as an out-of-distribution detection problem (Zeng et al., 2025). Instead of modeling human writing as one coherent distribution, HTAO treats machine-generated text as in-distribution and human-written text as out-of-distribution. This is useful because human writing varies widely across authors and domains, while LLM outputs can share more stable statistical patterns. HTAO evaluates one-class and energy-based methods such as DeepSVDD, HRN, and energy scoring. Our project builds on this OOD framing, but extends the problem to unknown LLM rejection and known-generator attribution.

2.3 Open-Set and Unknown-Aware Detection

Unknown-aware object detection studies how a model can detect classes that were not labeled during training. STUD, for example, uses unlabeled videos in the wild to learn what the model does not know, then applies energy-based regularization to separate known and unknown objects (Du et al., 2022). Although our final implementation does not directly use spatial-temporal distillation, this line of work motivated our open-set formulation and our focus on rejecting unknown generator families rather than forcing every LLM output into a known class.

2.4 Energy-Based OOD Detection

Energy-based OOD detection uses an energy score rather than only softmax confidence to identify samples that do not fit the training distribution (Liu et al., 2021). In our experiments, energy was one of the most useful signals for unknown LLM rejection. A direct three-way classifier confused known and unknown LLMs too often, while an energy threshold gave a more controllable tradeoff between detecting unknowns and preserving known LLM attribution.

2.5 Style Embeddings

Because LLM attribution often depends on writing style rather than topic, we used frozen StyleDistance embeddings as our representation backbone (Patel et al., 2025). StyleDistance is designed to capture content-independent stylistic similarity, making it a natural fit for distinguishing human writing, known LLM outputs, and unknown LLM outputs. However, our results also show that frozen style embeddings are limited under domain-balanced evaluation, motivating future fine-tuning.

3 Task Formulation

Given an input text x, the model must predict one of three high-level categories:

- **Human**: the text is human-written.
- **Known LLM**: the text is generated by one of the LLM families seen during training.
- **Unknown LLM**: the text is generated by an LLM family held out from training.

If the input is predicted as known LLM, the model also predicts the specific generator family. This makes the task both a detection problem and an attribution problem. The system must avoid two types of errors: misclassifying human writing as machine-generated, and forcing unknown LLM outputs into known generator labels.

4 Datasets

4.1 RAID

Our main experiments use RAID (Dugan et al., 2024), which contains human and LLM-generated text across multiple generators, domains, decoding strategies, and adversarial attacks. RAID matched our task better than the initial OpenTuringBench setup because it contains human examples as well as outputs from many LLM families. This allowed us to train and evaluate a full human, known LLM, and unknown LLM pipeline.

We ran two main RAID settings. The first was an abstracts-only setting, which produced the best model performance. The second was a domain-balanced setting using examples from abstracts, books, news, poetry, recipes, Reddit, reviews, and Wikipedia. The domain-balanced setup is more realistic because it tests whether the detector generalizes across writing styles and subject areas, but it was also more difficult.

4.2 OpenTuringBench

We initially tested an OpenTuringBench-style (Cava and Tagarelli, 2025) attribution setup. This was useful for early known-generator classification experiments, but the split we used contained no human examples in train, validation, or test. As a result, it could not support the full human-vs-known-vs-unknown task. We therefore switched to RAID for the main experiments.

4.3 DeepFake

The original project proposal also discussed DeepFake (Li et al., 2024), which contains human and machine-generated text from diverse sources. In the final project, DeepFake served mainly as background and motivation from prior work; our implemented experiments and reported results focus on RAID.

5 Methodology

5.1 Embedding Backbone

All models use frozen StyleDistance embeddings. The purpose of these embeddings is to represent writing style while reducing dependence on topic. We chose this representation because generator attribution often relies on stylistic patterns, such as phrasing, formatting, and distributional regularities, rather than the semantic content of the text.

Using frozen embeddings made the project computationally simpler and allowed us to test multiple classifiers and OOD scores quickly. However, it also became the main limitation of the final system. Since the embedding model was not fine-tuned for our exact human, known LLM, and unknown LLM objective, it did not always separate unknown generators cleanly from known ones.

5.2 Frozen-Embedding Baselines

The first full RAID baseline used frozen StyleDistance embeddings with three components:

- a human-vs-LLM logistic regression classifier,
- a known-LLM logistic regression classifier,
- OOD scores such as nearest-centroid distance, DeepSVDD-style distance, and energy.

This baseline made clear that human-vs-LLM detection was relatively strong and known-generator classification was also reasonable. The difficult part was unknown rejection. Early unknown gates worked on validation unknowns but did not generalize reliably to different held-out unknowns.

5.3 HTAO-Inspired OOD Methods

We then tested OOD methods inspired by HTAO, including DeepSVDD-style distance from an LLM embedding center, class-conditional DeepSVDD, and energy-based scoring. The OOD framing was helpful, but the unknown LLM setting was harder than human-vs-LLM detection. Unknown LLM outputs often remained close to known LLM outputs in embedding space, so simple distance-based rejection was unstable.

5.4 Ensemble Rejector

To improve unknown rejection, we trained an ensemble rejector using multiple signals: nearest-centroid distance, DeepSVDD distance, energy score, softmax confidence, softmax entropy, and softmax margin. A small logistic regression model learned to combine these features and reject unknown LLMs.

This improved unknown detection compared with using a single OOD score, but it was too aggressive. In one run, it detected about 64% of unknowns but falsely rejected about 37% of known LLMs. This false unknown rate was too high for a practical attribution system because many known LLM outputs were incorrectly labeled unknown.

5.5 Direct Three-Way Triage Model

We next tried a direct stage-one classifier with three labels: human, known LLM, and unknown LLM. If the text was classified as known LLM, a second classifier predicted the generator family. This matched the task formulation directly and was conceptually clean.

In practice, it performed worse than the cascade. The direct three-way model confused known and unknown LLMs too often, which prevented the stronger known-LLM classifier from being used. This result suggested that known-vs-unknown rejection should be handled by a separate thresholded OOD gate rather than a single multiclass classifier.

5.6 Final Cascade with Energy Unknown Gate

Given input text x, the shared encoder produces a frozen StyleDistance representation $h = \varphi(x) \in \mathbb{R}^d$. The cascade then makes three sequential decisions:

1. **Stage 1 (human-vs-LLM).** A logistic regression classifier estimates $p(\text{human} \mid h)$. If this probability exceeds a threshold τ_h, the system returns Human.

2. **Stage 2 (unknown gate).** Otherwise, the energy score $E(h)$ is computed from the known-LLM classifier's logits. If $E(h) > \tau_e$, the input is rejected as an Unknown LLM.

3. **Stage 3 (attribution).** If the input passes both gates, a K-way classifier predicts $\hat{y} = \arg\max_k f_k(h)$ over the seven known LLM families.

Figure 2 shows the full pipeline. This design worked best because each module solves a narrower problem: the human detector preserves high human recall, the energy gate controls the known-unknown tradeoff via the threshold τ_e, and the final classifier focuses only on attribution among known LLM families.

5.7 Experimental Settings

Dataset and splits. We used the RAID `train` and `extra` splits with twelve target labels: human, chatgpt, gpt4, gpt3, gpt2, llama-chat, cohere, cohere-chat, mpt, mpt-chat, mistral, and mistral-chat. The seven known LLM classes for attribution were chatgpt, gpt3, gpt4, cohere, llama-chat, mpt, and mpt-chat. We treated gpt2 and cohere-chat as wild unknown models for tuning the unknown gate, mistral as the validation unknown, and mistral-chat as the held-out test unknown. Adversarial RAID examples were included, empty texts were removed, and model names were normalized. Examples were grouped by `source_id` before splitting to reduce leakage, then split 70/15/15 by source group.

Sampling settings. We evaluated two sampling settings, both capped at 20,000 examples per label. In the abstracts-only setting, examples were collected by streaming RAID until per-label quotas were met; because RAID's stream order surfaces abstracts first, the resulting set was abstracts-dominated despite no explicit domain filter. In the domain-balanced setting, the per-label quota was divided evenly across eight RAID domains (abstracts, books, news, poetry, recipes, Reddit, reviews, and Wikipedia), giving 2,500 examples per label–domain pair while preserving the same target size.

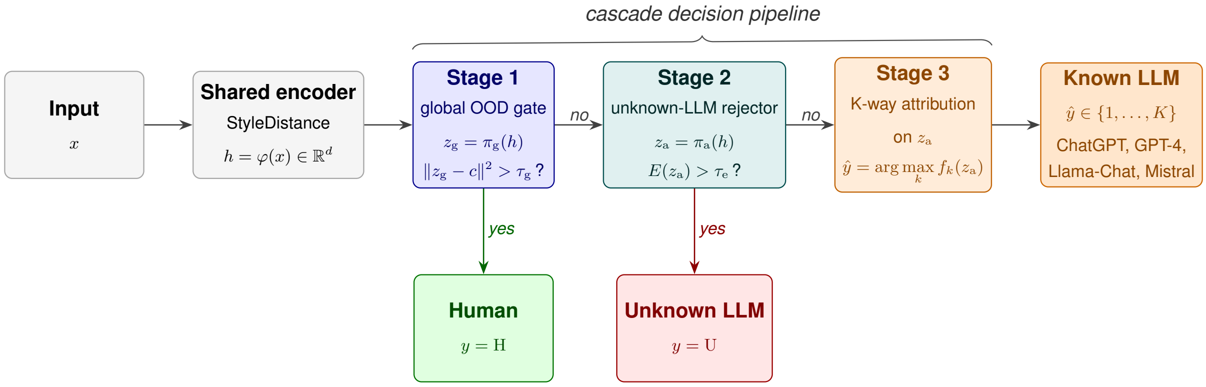

Figure 2: Final cascade decision pipeline. Stage 1 is a global OOD gate, Stage 2 an unknown-LLM rejector based on energy scoring, and Stage 3 the K-way attribution classifier.

Encoder. We used the frozen `StyleDistance/styledistance` encoder with no task-specific fine-tuning. Texts were tokenized with padding and truncation to 512 tokens. Embeddings were extracted from the second-to-last hidden layer and mean-pooled over non-padding tokens using the attention mask. We used a batch size of 32, fp16 autocasting on GPU, and fixed the random seed to 42.

Embedding extraction. StyleDistance is used as a frozen encoder. Inputs are tokenized with the model's default tokenizer at a maximum length of 512 tokens. Embeddings are produced by attention-mask-weighted mean pooling over the second-to-last hidden layer, in FP16 mixed precision, with batch size 32 on a single CUDA GPU.

Models compared. We used standardized, class-balanced logistic regression for both human-vs-LLM detection and known-LLM attribution. For unknown rejection, we compared centroid distance, class-conditional DeepSVDD distance, energy scoring (Liu et al., 2021), confidence-based scores, and a logistic-regression ensemble over eight OOD features. DeepSVDD radii used the 95th percentile of class-conditional cosine distances, with energy temperature set to 1.0.

Final cascade. The human-vs-LLM classifier first predicted whether an input was human-written or machine-generated; human predictions were returned as human. LLM-predicted inputs were passed through an energy-based unknown gate: if the energy score exceeded the threshold, the input was labeled as an unknown LLM; otherwise, the known-LLM classifier predicted one of the seven known generators.

Threshold Selection. Energy gate thresholds were chosen on the wild-unknown validation split. The known and unknown validation scores were pooled, 400 quantile-spaced values were selected and the candidate maximizing $0.5 \cdot (\text{known kept}) + 0.5 \cdot (\text{unknown rejected})$ subject to a known-false-unknown cap. We tested caps of 0.10, 0.15, and 0.18 in the abstracts-only setting, and used 0.18 for both final cascades because it gave the best tradeoff between unknown detection and known attribution.

Compute. All experiments ran on a single NVIDIA L40 GPU.

6 Evaluation Metrics

We report the following metrics:

- **Final cascade accuracy**: overall accuracy after the full cascade.
- **Human recall**: percentage of human examples correctly returned as human.
- **Unknown detection rate**: percentage of unknown LLM examples correctly flagged as unknown.
- **Unknown precision, recall, and F1**: precision, recall, and F1 for the unknown class.
- **Known false unknown rate**: percentage of known LLM examples incorrectly rejected as unknown.

- **Known attribution accuracy after gate**: accuracy of the known-LLM classifier on examples that pass the unknown gate.

These metrics are necessary because a high overall accuracy can hide poor unknown rejection, while a high unknown detection rate can be achieved by rejecting too many known LLMs. The key tradeoff is detecting unknown LLMs without making the false unknown rate too large.

7 Results

7.1 Threshold Tuning

For the abstracts-only cascade, we tuned the energy threshold to balance unknown detection against false unknown rejections. Table 1 summarizes the main threshold settings.

The 0.10 threshold was conservative. It protected known LLMs, with only 8.37% known false unknown rate, but missed too many unknowns. The 0.15 threshold was a better middle ground. The 0.18 threshold gave the strongest overall result, with 72% final accuracy, 61.79% unknown detection rate, 0.69 unknown F1, and 93.60% known attribution accuracy after the gate.

7.2 Best Abstracts-Only Cascade

The best-performing model was the abstracts-only cascade with an energy threshold of 0.18. It achieved:

- 72% final cascade accuracy,
- 79% unknown precision,
- 62% unknown recall,
- 0.69 unknown F1,
- 15.42% known false unknown rate,
- 61.79% unknown detection rate,
- 93.60% known attribution accuracy after the gate,
- 97.03% human recall.

These results show that the cascade successfully separates the problem into useful subproblems. Human detection is strong, known-generator attribution remains strong after the gate, and unknown detection is meaningfully better than the earlier conservative baselines. The main weakness is that unknown recall is still only about 62%, meaning many unknown LLM examples are still assigned to known generator families.

7.3 Domain-Balanced Cascade

We then tested a domain-balanced version using examples from abstracts, books, news, poetry, recipes, Reddit, reviews, and Wikipedia. The goal was to reduce reliance on abstract-specific artifacts and evaluate a more realistic setting.

Table 2 compares the domain-balanced cascade to the best abstracts-only cascade. Human recall remained high, but unknown detection and final accuracy dropped substantially. This suggests that the frozen embedding model may be learning domain-specific cues in the abstracts-only setting and does not fully generalize when domain variation increases.

8 Discussion

8.1 Why the Cascade Worked Best

The cascade worked better than a direct three-way model because the subtasks have different difficulty levels. Human-vs-LLM detection is easier than unknown LLM rejection, and known-generator attribution is relatively strong once the model has already decided the input belongs to a known LLM. A direct three-way classifier must learn all boundaries at once, which caused it to confuse known and unknown LLMs. The energy gate gave us a tunable decision rule for this hardest boundary.

8.2 Unknown Detection Remains the Main Weakness

Even in the best setting, unknown recall was only about 62%. This is the central challenge of the project. Unknown LLMs are not obviously out-of-distribution in the same way human writing is. They often share stylistic patterns with known LLMs, especially if model families are related or trained on similar data. This makes unknown rejection much harder than standard human-vs-machine detection.

8.3 Domain Shift Hurts Performance

The drop from 72% accuracy in the abstracts-only setting to 62% in the domain-balanced setting shows that domain shift is a major issue. A detector may perform well when training and testing text come from the same domain, but struggle when the writing style changes because of genre rather than generator identity. This is especially important for real deployment, where text can come from academic abstracts, social media, news, reviews, recipes, and many other domains.

Energy threshold	Final acc.	Unknown det.	Unknown F1	Known false unknown	Known attr. after gate
0.10	67%	44.74%	0.58	8.37%	92.37%
0.15	70%	56.09%	0.66	12.45%	93.16%
0.18	**72%**	**61.79%**	**0.69**	15.42%	**93.60%**

Table 1: Energy threshold tuning for the abstracts-only cascade. Larger thresholds improve unknown detection but also increase false unknown rejections among known LLMs.

Model	Final acc.	Unknown det.	Unknown F1	Known false unknown	Human recall
Abstracts-only cascade, 0.18 gate	**72%**	**61.79%**	**0.69**	15.42%	**97.03%**
Domain-balanced cascade	62%	45.87%	0.56	18.00%	96.77%

Table 2: Comparison between the best abstracts-only cascade and the domain-balanced cascade. The domain-balanced setting is more realistic but substantially harder.

8.4 Limitations

The biggest limitation is that the embedding model was frozen. Frozen StyleDistance embeddings provided useful style features, but they were not optimized for our exact open-set objective. A second limitation is that the energy gate depends on threshold tuning. Although thresholding gives useful control over the unknown detection tradeoff, the best threshold may change across domains or generator families. A third limitation is that all experiments were conducted on the RAID dataset, so it is unclear how well the findings transfer to text from other sources, domains, or generator distributions not represented in RAID. Finally, the current model is not robust enough under domain-balanced evaluation, which means it should be viewed as a strong baseline rather than a finished detector.

9 Future Work

The next improvement should be representation learning. Instead of using frozen embeddings, future work should fine-tune the encoder with a contrastive or metric-learning objective that explicitly separates human text, known LLMs, and held-out LLM families. Another direction is to train with more diverse wild data so the unknown gate sees a broader range of generator and domain shifts. Finally, threshold calibration should be studied more carefully so the system can maintain a stable false unknown rate across domains.

10 Conclusion

We built an unknown-aware LLM-generated text detection system that distinguishes between human-written text, known LLM outputs, and unknown LLM outputs. Our best model used frozen StyleDistance embeddings, a human-vs-LLM classifier, an energy-based unknown gate, and a known-LLM classifier. The best abstracts-only model achieved 72% final accuracy, 97.03% human recall, 61.79% unknown detection rate, and 93.60% known attribution accuracy after the gate. Performance dropped in the domain-balanced setting, showing that domain shift and unknown LLM rejection remain difficult. Overall, energy-based unknown gating is promising, but future work should focus on fine-tuned representations and stronger domain generalization.

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